Lab 3.3 Discipline specific specification pack

IES | 20 minutes | LO1, LO2, LO3

All records are fictional training data. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use outputs to operate, maintain or approve real equipment.

Inputs and preparation

Open the mock-data folder: discipline-options.csv, spec-pack-template.md, DES-01-rev2.md. Use prompts.md for copy and paste. Save your reviewed work under outputs. Earlier lab outputs are optional; this folder provides standalone inputs.

Procedure

1. Choose a discipline from discipline-options.csv. Use the supplied fictional parameters; do not upload actual employer documents.

2. Create four sections or files from spec-pack-template.md: specification, design checklist, SOP planning outline and troubleshooting guide. State the chosen component and missing domain inputs.

3. Prompt Copilot to draft each component separately so the checklist can be traced to the specification. For non-pump domains, DES-01 is not an applicable design authority; retain relevant domain gaps instead.

Lab 3.3 continued

4. Add requirement IDs and evidence fields to the checklist. Keep SOP content at planning level unless approved operating procedures are supplied. Troubleshooting recommendations should identify evidence and escalation.

5. Exchange the pack with a neighbour. Score completeness, traceability, uncertainty handling and safe review status from 0-2 each. Correct one issue and submit the four-component pack to the mini challenge.

Check before moving on

All four components are present and consistent.

Domain-specific gaps are retained and irrelevant pump criteria are excluded.

Access or source problems

Use approved paired access or the local source files. Record unavailable live capabilities as not executed. If an attachment is not supported, paste the short synthetic text or reference its approved SharePoint copy. Do not use personal accounts to bypass controls.

Lab 3.3 prompts and evidence

Main prompt

Act as a documentation assistant for the selected discipline in discipline-options.csv. Produce the four components in spec-pack-template.md. Use only relevant supplied parameters, retain all unknowns, and cross-reference requirements and checks. Do not invent standards or operating instructions. Include a reviewer gate and evidence requests.

Verification prompt

Audit the draft against the named source files. List each claim, source location, uncertainty and correction. Recalculate key values independently. Return a revised draft and change log; do not approve the engineering deliverable.

Expected output example

Illustrative teaching example, not a recorded AI run. Electrical example: two pump status inputs are known; supply and protection ratings remain TBD. Checklist asks for approved interface requirements. Troubleshooting asks for evidence of signal state and authorised escalation.

Retain your evidence

Keep the source versions, prompt, tool, output, corrections, independent checks and reviewer decision. Record any capability or permission test that was not executed. Use templates/prompt-log.csv. Do not publish real workplace data.